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Lab 1: Implementation of Source code management using git commands through github

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BESE2023

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OBJECTIVE:

To learn and practice Source Code Management (SCM) by using Git commands together with GitHub for version control and collaborative software development.

AIM:

To gain practical experience in managing source code with Git and GitHub, including tracking file modifications, maintaining version history, and supporting team-based development.

THEORY:

Source Code Management (SCM) is a software development practice that manages and records changes made to source code over time. It allows developers to monitor modifications, restore earlier versions of code, and coordinate work efficiently in multi-developer environments.

Git is an open-source distributed version control system that helps developers track changes in project files, create different versions of the code, and merge contributions from multiple team members without losing previous work.

GitHub is an online platform that hosts Git repositories and provides collaboration features such as repository sharing, pull requests, issue management, and project organization. It enables developers to work together from different locations while maintaining a centralized record of the project's progress.

The integration of Git and GitHub plays an important role in Agile Software Development by supporting efficient collaboration, version management, continuous integration, code reviews, and organized software development workflows.

OBJECTIVES:

- Learn the basics of Git.
- Create and manage Git repositories.
- Track project changes using commits.
- Connect local repositories with GitHub.
- Push and pull project updates.
- Understand collaborative software development.

ALGORITHM:

1. Install Git on the system.
2. Create a GitHub account.
3. Create a repository on GitHub.
4. Initialize a local Git repository.
5. Add project files.
6. Stage files using Git.
7. Commit changes.
8. Link local repository to GitHub.
9. Push files to GitHub.
10. Verify successful synchronization.

CODE:

- Initialize Repository

```
git init
```

- Configure User Information

```
git config --global user.name "your name"
git config --global user.email your@email.com
```

- Check Repository Status

```
git status
```

- Add Files

```
git add .
```

- Commit Changes

```
git commit -m "Initial Commit"
```

- Connect Repository to GitHub

```
git remote add origin https://github.com/username/repository.git
```

- Push to GitHub

```
git branch -M main
git push -u origin main
```

- Pull Latest Changes

```
git pull origin main
```

OUTPUT:

The Git repository was successfully initialized and connected to GitHub.

- Repository created successfully.
- Files tracked using Git.
- Changes committed successfully.
- Remote repository linked.
- Project pushed to GitHub.

RESULT:

The Source Code Management tasks were carried out successfully using Git and GitHub. Essential version control operations, including repository initialization, adding files, creating commits, and pushing changes to a remote GitHub repository, were completed without issues.

CONCLUSION:

This experiment enhanced the understanding of Source Code Management through hands-on use of Git and GitHub. It highlighted the importance of version control in tracking code changes, simplifying project management, facilitating team collaboration, and supporting efficient software development within Agile environments.